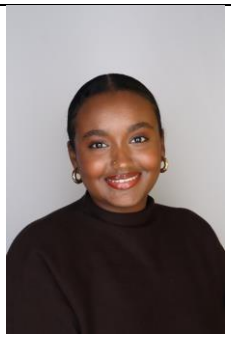
SP-108 Green ASM

CS 4850 - Section 01 – Spring 2026

1/25/2026

Project Team

Roles	Name	Role	Contact (Cell Phone)
Team leader	Alex Tidwell	Documentation + Developer (as needed)	678-773-1588
Team members	Katherine Smith	Documentation + Developer	561-542-8609
	Philip Eiser	Developer	470-526-1204
	Brianna Lewis	Developer	470-562-8166
	Sumaya Tamenne	Documentation	404-944-9608
Advisor / Instructor	Sharon Perry	Facilitate project progress; advise on project planning and management.	770-329-3895

				
Alex Tidwell Team Leader Documentation + Developer (as needed)	Katherine Smith Documentation + Developer	Sumaya Tamenne Documentation	Phillip Eiser Developer	Brianna Lewis Developer

1.0 Project Overview

This project consists of designing and creating an Attack Surface Management (ASM) system used for the tracking and monitoring of what an individual or organization is exposing to the world. Instead of starting with a prespecified list of assets, the project approaches the environment from an outside perspective. The ASM will be designed to focus on discovering what is reachable by unauthorized users or scammers including systems that may be undocumented, forgotten, or unintentionally exposed.

The overall aim of this project is to create an automated ASM framework using various open-source tools. We intend to design the management system so that it continuously identifies domains, subdomains, hosts, and services. The system will then associate those assets with known vulnerabilities and highlight changes in the attack surface overtime. A key part of this project is that the ASM will be repeatable, constantly updating to ensure the protection of assets. This all ties together and allows the environment to be monitored even as it evolves, rather than treated as a snapshot.

Project Website

We decided to name our website www.spideysensesASM.com because like "spidey senses" our ASM engine will help detect possible threats.

Deliverables

1. **Software Requirements Specification and Software Design Document**
2. **The ASM Engine:** A fully functional, containerized stack that can scan a domain/IP range and output a report.
3. **Asset Inventory Log:** A live-updated list (CSV, Database, or GUI) showing all discovered assets, their status, and risk level.
4. **"Drift" Reports:** A weekly report highlighting changes in the attack surface (e.g., A new port 8080 was opened on Server B yesterday").
5. **Software Test Plan and Software Test Report**
6. **Final Technical Paper:** A 10-page report detailing the architecture, tool selection rationale, and a "post-mortem" of the discovered vulnerabilities.

Group Meeting Schedule Date/Time

We plan to meet weekly on Tuesdays from 3 p.m. to 4:30 p.m.

Collaboration and Communication Plan

We will meet weekly on campus and have any extra clarification meetings via Teams or Discord messaging.

Project Schedule and Task Planning

Project Name: SP-108 Green ASM
Report Date: 1/25/2026

						Phase #1		*Phase #2*		*Phase #3*			*Phase #4*			*Red Phase*	Final Phase			
Phases	**Deliverables**	**Tasks**	**Complete%**	**Current Status Memo**	**Assigned To**	**01/25**	**02/01**	**02/08**	**02/15**	**02/22**	**03/01**	**03/08**	**03/15**	**03/22**	**03/29**	**04/05**	**04/12**	**04/19**	**04/26**	**05/03**
Foundations		Create GitHub version control	0%		Katherine	1														
		Review tutorials for platforms	0%		Katherine, Brianna, Phillip	10														
		Research and set up a cloud/network environment via Docker	0%		Sumaya	3	2													
		Research and set up a cloud/network environment via VM	0%		Alex	3	2													
		Create SRS and SDD	0%		Alex and Sumaya	1	4													
	SRS and SDD	Create a local DNS server to simulate a corporate domain/subdomains	0%		Katherine, Brianna, Phillip		7													
Manual Discovery		Study tools for manual reconnaissance	0%		Katherine, Brianna, Phillip			7												
		Perform manual reconnaissance	0%		Everyone			3	3											
		Map the initial attack surface	0%		Everyone				5											
		Create environment for ASM engine	0%		Katherine, Brianna, Phillip				7											
Automation & Vulnerability Layer		Integrate tools into the ASM engine	0%		Katherine, Brianna, Phillip					10	5									
		Correlate assets with known CVEs	0%		Everyone					2										
		Write scripts to run scans weekly	0%		Katherine, Brianna, Phillip						5									
		Start creating Drift Reports	0%		Everyone						2	2								
	ASM Engine	Set up database to store results of scans	0%		Everyone						2	8								
Shadow IT Simulation & UI		Create GUI dashboard to view reports	0%		Everyone								10							
		Add "unexpected" assets to the environment to test the ASM engine	0%		Katherine, Brianna, Phillip								4	4						
	Custom Dashboard	Add Asset Inventory Log to GUI	0%		Katherine, Brianna, Phillip									7						
	STP and STR	Create STP and STR	0%		Alex and Sumaya									4	6					
Red Phase Testing		Evasion testing	0%		Everyone											10				
		Protocol tunneling	0%		Everyone											10				
		Vulnerability obfuscation	0%		Everyone											10				
		Take note of issues that come up	0%		Everyone											5				
Final Delivery		Perform "Purple Phase" improvements to update the gaps in the ASM engine	0%		Katherine, Brianna, Phillip												8	8	4	
		Alternate red phase testing and engine improvements	0%		Everyone												6	6	3	
		Create presentation video	0%		Alex														2	
		Create Final Report, including "Drift Reports" (weekly report highlighting changes in the attack surface)	0%		Alex, Sumaya, and Katherine														4	
		Create Website	0%		Everyone													3	3	
	Executive briefing	Final report submission to D2L and project owner	0%		Everyone														1	
				Total work hours	**211**	**17**	**15**	**10**	**15**	**12**	**14**	**10**	**14**	**15**	**6**	**35**	**14**	**17**	**17**	**0**

* formally define how you will develop this project including source code management

Legend	
Planned	
Delayed	
Number	Work: man hours

Other Plans

Since our group is a 5-person team, we plan on adding a separate cloud container or VM, in addition to the network environment that is built. Adding the cloud environment will require extra time to learn the tools we need to create, maintain, and secure the environment, as well as the different attack surface that comes along with it.

Version Control Plan

We will be using GitHub for version control. We will have a group GitHub for the project created as an organization. Everyone will be able to contribute as an individual contributor to work on the project within the organization GitHub account.